MATH73235 Spring 2025 Quiz 6

First name:

Last name:

Student ID:

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Problem 1. *Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function $f(x,y,z) = x^2 + y^2 + z^2$ in both Cylindrical and Spherical coordinates. You must compute all the required partial derivatives.*

For generic coordinates (q_1, q_2, q_3):

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial q_1}\left(\frac{h_2 h_3}{h_1}\frac{\partial f}{\partial q_1}\right) + \frac{\partial}{\partial q_2}\left(\frac{h_3 h_1}{h_2}\frac{\partial f}{\partial q_2}\right) + \frac{\partial}{\partial q_3}\left(\frac{h_1 h_2}{h_3}\frac{\partial f}{\partial q_3}\right)\right].$$

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

For spheical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin\phi, \quad h_3 = h_\phi = \rho.$$

[5 + 5 = 10 marks]

Solution.

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Extra Page: Must submit even if unused.

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